

Adaptive Robust Control of Wheel-Legged Self-Balancing Robot Based on LQR

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Abstract: This paper develops an adaptive linear quadratic regulator control framework for wheel-legged self-balancing robots, employing a double inverted pendulum model with Jacobian linearization to address variable leg-length dynamics (0.1 – 0.4 m). The controller integrates quadratic gain scheduling and virtual model control, achieving ± 0.006 rad chassis stabilization and $\leq 2\%$ velocity tracking error in simulations. The hierarchical architecture coordinates leg-length adjustments and steering dynamics through hybrid PID-LQR strategies, demonstrating enhanced robustness against parametric variations compared to fixed-gain methods.

Key Words: Wheel-legged robot, Adaptive robust control, Linear quadratic regulator (LQR), Double inverted pendulum, Gain scheduling

1 Introduction

The robotics industry has witnessed remarkable advancements in recent decades, with wheeled and legged robots emerging as two dominant paradigms in mobile robotics[1]. Wheel-legged robots, as innovative hybrid platforms combining the advantages of both locomotion modalities, demonstrate exceptional potential in terrain adaptability, motion efficiency, and operational versatility[2]. This dual-capability architecture enables Wheel-legged robots to navigate complex environments ranging from urban infrastructures to unstructured outdoor terrains, making them particularly suitable for service robotics, emergency response, and planetary exploration applications[3].

Recent advancements in Wheel-legged robot development have produced notable prototypes such as ETH Zurich's Ascento with its dynamic jumping capability[4], Boston Dynamics' Handle featuring advanced whole-body control[5], and Tencent's Ollie utilizing parallel five-bar mechanisms[6]. These implementations highlight two critical research directions: 1) mechanical design optimization for hybrid locomotion, and 2) sophisticated control strategies for dynamic stability. Despite these achievements, maintaining robust balance under variable leg configurations and external disturbances remains a fundamental challenge, particularly when addressing the underactuated nature of wheel-leg systems[3].

Contemporary balance control architectures for wheel-legged robots predominantly employ three algorithmic paradigms: cascaded PID control[7], model predictive control (MPC)[8], and linear quadratic regulator (LQR) control[9]. As demonstrated in prior studies, each approach exhibits distinct performance characteristics. While computationally efficient, cascaded PID control demonstrates limited disturbance rejection capability. As for MPC implementations, although the LQR Controller provides mathematically guaranteed stability through Riccati equation solutions, but employs fixed gain matrices that cannot accommodate the nonlinear parameter variations induced by leg-length adjustments (0.1-0.4 m operational range in our platform).

Retically optimal, it requires precise system identification and suffers from prohibitive computational complexity[10]. LQR Controller provides mathematically guaranteed stability through Riccati equation solutions, but employs fixed gain matrices that cannot accommodate the nonlinear parameter variations induced by leg-length adjustments (0.1-0.4 m operational range in our platform). To address these limitations, an adaptive robust LQR (AR-LQR) control has been proposed[11].

In this paper, the wheel-legged robotic system is abstracted into a double inverted pendulum (DIP) paradigm to enable systematic force analysis through Newtonian mechanics in section 2. This topological simplification preserves critical dynamic couplings while eliminating high-frequency complexities, establishing a tractable foundation for controller synthesis. Building upon the derived state-space representation, an adaptive robust linear quadratic regulator (AR-LQR) is formulated with two core innovations in section 3: 1) real-time gain matrix $\mathbf{K}$ adaptation via quadratic polynomial regression across discrete leg-length configurations (5 mm resolution, $L \in [0.1, 0.4]$ m), and 2) virtual model control integration for enhanced disturbance rejection. Theoretical efficacy is rigorously validated through Webots simulations, demonstrating simultaneous chassis stabilization (± 0.006 rad) and trajectory tracking (steady-state error $< 2\%$).

2 System Modeling and Analysis

The wheel-legged self-balancing robot inherently possesses hybrid nonlinear dynamics and underactuation characteristics[12], arising from the dual mobility mechanisms of wheels (for planar motion) and articulated legs (for terrain adaptation). To address the complexity of real-time control while preserving essential dynamic couplings, the system is abstracted into a DIP paradigm (Fig. 1)[13]. This reduced-order model strategically retains:

- Critical balancing states: Tilt angles of the chassis-leg-wheel chain,
- Dominant nonlinearities: Inertial coupling between ro-

tational joints and wheel-ground interactions,

- Actuation constraints: Limited torque authority from wheel motors to stabilize two passive rotational degrees-of-freedom.

By neglecting high-frequency leg flexure dynamics and assuming rigid-contact rolling constraints, the DIP approximation enables tractable controller synthesis while maintaining fidelity for tilt angles within $\pm 0.3 \text{ rad}$ —a critical operational regime for fall prevention.

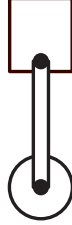


Fig. 1: DIP Model.

2.1 Wheel Model

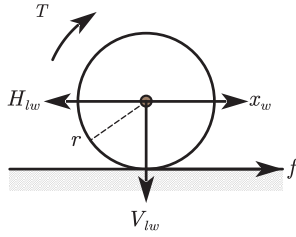


Fig. 2: Wheel Model.

Table 1: Physical Quantities in Wheel Model

Symbol	Meaning	Unit	Direction
T	Wheel drive motor torque	$N \cdot m$	Clockwise
f	Ground friction force on wheels	N	Horizontal rightward
H_{lw}	Horizontal leg force	N	Horizontal leftward
V_{lw}	Vertical leg force	N	Vertical downward
x_w	Wheel horizontal displacement	m	Horizontal rightward

By decoupling the wheel's translational and rotational dynamics, the following equations are derived:

$$m_w \ddot{x}_w = f - H_{lw}, \quad (1)$$

$$I_w \ddot{\theta}_w = Tr - fr^2, \quad (2)$$

where m_w (mass), I_w (moment of inertia), and r (radius) are the wheel's key inertial-geometric parameters governing its planar dynamics.

2.2 Leg Model

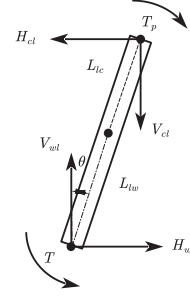


Fig. 3: Leg Model.

Table 2: Physical Quantities in Leg Model

Symbol	Meaning	Unit	Direction
θ	Leg tilt angle relative to vertical	rad	Clockwise
T_p	chassis actuation torque	$N \cdot m$	Clockwise
H_{wl}	Horizontal leg reaction force	N	Horizontal rightward
V_{wl}	Vertical leg reaction force	N	Vertical upward
H_{cl}	Horizontal chasis force	N	Horizontal leftward
V_{cl}	Vertical chasis force	N	Vertical downward

The governing equations for the leg's decoupled translational dynamics are derived as follows:

$$H_{wl} - H_{cl} = m_l(\ddot{x}_w + \frac{\partial^2}{\partial t^2}(L_{lw} \sin \theta)), \quad (3)$$

$$V_{wl} - V_{cl} - m_l g = m_l \frac{\partial^2}{\partial t^2}(L \cos \theta). \quad (4)$$

As for rotation:

$$I_l \ddot{\theta} = (V_{wl} L_{lw} + V_{cl} L_{lc}) \sin \theta + T_p - (H_{wl} L_{lw} + H_{cl} L_{lc}) \cos \theta - T, \quad (5)$$

the leg kinematic parameters in (3), (4) and (5) are defined as:

- L_{lw} : Distance between the leg's center of mass of the leg (CoM) and the rotor drive axis of the wheel, which governs the inertial coupling of the leg-wheel,
- L_{lc} : Distance from leg CoM to suspension motor axis, determining torque transmission efficiency.

The remaining terms represent the leg's mass(m_l), moment of inertia(I_l) and the gravitational acceleration(g).

2.3 Chassis Model

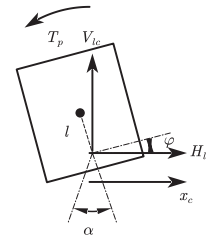


Fig. 4: Chassis Model.

Table 3: Physical Quantities in Chassis Model

Symbol	Meaning	Unit	Direction
α	Leg-chassis vertical tilt angle	rad	Clockwise
φ	Chassis pitch angle	rad	Counterclockwise
x_c	Chassis horizontal displacement	m	Horizontal rightward
H_{lc}	Horizontal chassis reaction force	N	Horizontal rightward
V_{lc}	Vertical chassis reaction force	N	Vertical upward

Employing a parallel derivation methodology to the leg dynamics, the chassis governing equations are formulated through Newton-Euler mechanics [14], preserving the state-space structure while incorporating inertial coupling terms from the suspended leg assembly:

$$H_{lc} = m_c(\ddot{x}_c - \frac{\partial^2}{\partial t^2}(l \sin \varphi)), \quad (6)$$

$$V_{lc} - m_c g = m_c \frac{\partial^2}{\partial t^2}((L_{lw} + L_{lc}) \cos \theta + l \cos \varphi), \quad (7)$$

$$I_c \ddot{\varphi} = T_p + H_{lc} l \cos \varphi + V_{lc} l \sin \varphi, \quad (8)$$

where m_c and I_c are physical parameters of chassis, and l represents the distance from the center of mass of the chassis to the axis of rotation.

2.4 DIP Model

Through systematic kinetic analysis of the wheel-leg-chassis assembly, three mutually coupled dynamical subsystems are rigorously established:

- **Wheel dynamics** considering rolling friction saturation and slip constraints,
- **Leg dynamics** incorporating articulated joint damping and inertial cross-coupling,
- **Chassis dynamics** with terrain-induced disturbance propagation.

The governing equations are then synthesized into a unified DIP paradigm through Newtonian mechanics. By defining the state vector $\mathbf{x} = [x_w \ x_w \ \theta \ \dot{\theta} \ \varphi \ \dot{\varphi}]^T$ and control input $\mathbf{u} = [T \ T_p]^T$, the system admits the following state-space representation:

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}, \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}. \end{cases} \quad (9)$$

The output equation is configured for full-state feedback control with:

$$\mathbf{y} = \mathbf{x},$$

where:

- $\mathbf{y} \in \mathbb{R}^6$: Measured output vector,
- $\mathbf{x} \in \mathbb{R}^6$: System state vector defined in (2.4).

This configuration corresponds to the canonical output matrices:

$$\mathbf{C} = \mathbf{I}_6 \quad (6 \times 6 \text{ identity matrix}),$$

$$\mathbf{D} = \mathbf{0}_{6 \times 2} \quad (\text{null matrix}).$$

The DIP system exhibits inherent nonlinearities in its dynamics due to trigonometric terms in the Lagrangian and velocity-squared Coriolis components. To enable linear control synthesis, we perform Jacobian-based local linearization about the unstable equilibrium point $\mathbf{x}_0 = [x_w \ 0 \ 0 \ 0 \ 0 \ 0]^T$ (upright position):

$$\begin{aligned} \dot{\mathbf{x}} &\approx \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_{\mathbf{x}_0, \mathbf{u}_0} \delta \mathbf{x} + \left. \frac{\partial \mathbf{f}}{\partial \mathbf{u}} \right|_{\mathbf{x}_0, \mathbf{u}_0} \delta \mathbf{u} \\ &= \mathbf{A} \delta \mathbf{x} + \mathbf{B} \delta \mathbf{u}, \end{aligned} \quad (10)$$

where:

- $\delta \mathbf{x} = \mathbf{x} - \mathbf{x}_0$: State deviation from equilibrium,
- $\delta \mathbf{u} = \mathbf{u} - \mathbf{u}_0$: Control variation ($\mathbf{u}_0$: gravity compensation torque),
- $\mathbf{A} \in \mathbb{R}^{6 \times 6}$: Jacobian of system dynamics w.r.t states,
- $\mathbf{B} \in \mathbb{R}^{6 \times 2}$: Jacobian w.r.t. control inputs.

The linearization validity condition requires small angular deviations.

The symbolic Jacobian matrices $\mathbf{A}$ and $\mathbf{B}$ derived in (10) are parametrized with the robot's physical constants. Through MATLAB's *Symbolic Math Toolbox*, it is implemented:

$$\mathbf{A} = \begin{bmatrix} 0 & 1.0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -12.0703 & 0 & -0.8228 & 0 \\ 0 & 0 & 0 & 1.0 & 0 & 0 \\ 0 & 0 & 252.1919 & 0 & 82.2436 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1.0 \\ 0 & 0 & 342.5071 & 0 & 294.2640 & 0 \end{bmatrix}, \quad (11)$$

$$\mathbf{B} = \begin{bmatrix} 0 & 0 \\ 10.233 & -4.3104 \\ 0 & 0 \\ -125.6410 & 165.4161 \\ 0 & 0 \\ -111.0216 & 436.1361 \end{bmatrix}. \quad (12)$$

This algorithmic approach:

- Circumvents error-prone manual derivation of a lot of partial derivative terms in $\mathbf{A}$ and $\mathbf{B}$,
- Enforces dimensional homogeneity via automated unit consistency checks,
- Facilitates rapid reconfiguration for different robot morphologies.

2.5 Controllability Analysis

To verify the feasibility of linear state feedback control, we perform a controllability assessment of the linearized system $(\mathbf{A}, \mathbf{B})$. According to Kalman's rank condition [15], the system is fully controllable if:

$$\text{rank}(\mathcal{C}) = \text{rank} \begin{bmatrix} \mathbf{B} & \mathbf{A}\mathbf{B} & \mathbf{A}^2\mathbf{B} & \cdots & \mathbf{A}^5\mathbf{B} \end{bmatrix} = 6. \quad (13)$$

The controllability of the system is mathematically confirmed as Equation (13) is rigorously validated through MATLAB-based numerical computations.

3 Self-balancing controller design

3.1 LQR Controller

The linear quadratic regulator (LQR) [16] is synthesized for the linearized system $(\mathbf{A}, \mathbf{B})$ from Section 2.4. The opti-

mal control law minimizes the quadratic cost function:

$$J = \int_0^\infty (\mathbf{x}^T(t)Q\mathbf{x}(t) + \mathbf{u}^T(t)R\mathbf{u}(t)) dt. \quad (14)$$

3.1.1 Weighting Matrix Selection

The state weighting matrix $Q \in \mathbb{R}^{6 \times 6}$ and control weighting matrix $R \in \mathbb{R}^{2 \times 2}$ are designed through:

$$\begin{aligned} Q &= \text{diag}([500 \ 100 \ 1 \ 1 \ 1000 \ 1]), \\ R &= \text{diag}([1 \ 0.25]). \end{aligned} \quad (15)$$

The weighting matrices Q and R are designed through a systematic integration of Bryson's normalization principle and stability-critical state prioritization, ensuring alignment with physical system constraints and dynamic performance requirements. By normalizing weights to maximum allowable deviations and actuator limits—such as assigning $Q_{11} = 1/(0.035)^2 \approx 1000$ for the chassis pitch angle φ ($\Delta\varphi_{max} = 2^\circ \approx 0.035 \text{ rad}$). Priority is hierarchically allocated to stability-governing states: maximum weight on φ to prevent tipping, moderate weight on wheel position x_w (500) for trajectory tracking fidelity, and minimal weights on angular velocities (1) to suppress high-frequency oscillations, as validated through Bode sensitivity analysis.

The state feedback gain $K \in \mathbb{R}^{2 \times 6}$ is obtained by solving the algebraic Riccati equation:

$$A^T P + P A - P B R^{-1} B^T P + Q = 0, \quad (16)$$

yielding:

$$K = R^{-1} B^T P. \quad (17)$$

3.1.2 Optimal Gain Computation

The optimal feedback gain matrix K is systematically determined through MATLAB's Control System Toolbox via numerical solution of the algebraic Riccati equation.

3.1.3 Simulation verification

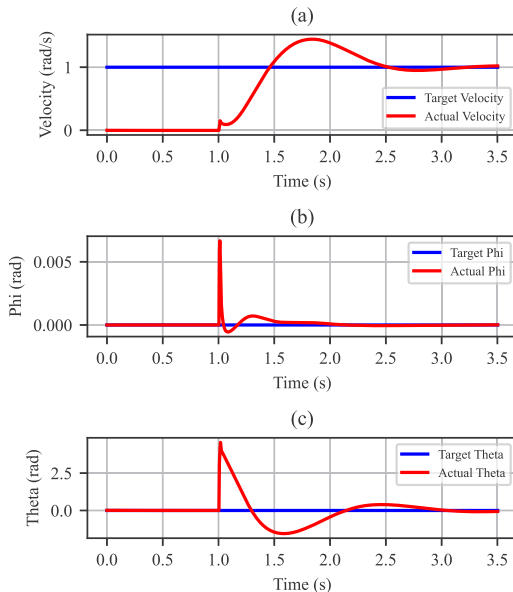


Fig. 5: Simulation Environment Validation Results.

As shown in Fig. 5, the system achieves precise tracking of the target angular velocity (1 rad/s) in Webots while maintaining chassis stability:

- **Steady-State Achievement:** The actual speed reaches within a $\pm 2\%$ error margin of the target speed in approximately 2.1 seconds, as shown in Fig. 5 (a).
- **Chassis Control Limitation:** The pitch angle is restricted to $\varphi_{max} = 0.006 \text{ rad}$, which is well below the 0.0175 rad safety threshold, ensuring safety and stability as depicted in Fig. 5 (b).
- **Leg Orientation Convergence:** The leg inclination angle θ exhibits asymptotic convergence to vertical alignment ($\theta \rightarrow 0 \text{ rad}$) using locomotion, achieving steady-state orientation control with $|\theta| < 0.002 \text{ rad}$ after 2.8 seconds in Fig. 5 (c).

3.2 Adaptive Robust Control

The variable-length wheel-leg mechanism is kinematically abstracted as a five-linkage model (Fig. 6), where l_0 denotes the effective leg length in the DIP configuration.

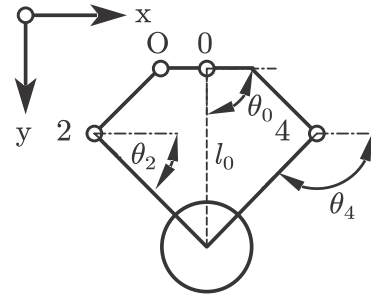


Fig. 6: Five-Link Diagram.

3.2.1 Five-link diagram

According to Fig. 6:

$$\begin{aligned} x_2 + l_2 \cos \theta_2 &= x_4 + l_2 \cos \theta_4, \\ y_2 + l_2 \sin \theta_2 &= y_4 + l_2 \sin \theta_4, \end{aligned} \quad (18)$$

the solution to the governing equation 18 is derived through rigorous algebraic manipulation, yielding:

$$\theta_2 = \frac{B_0 + \sqrt{A_0^2 + B_0^2 - l_{24}^4}}{A_0 + l_{24}^2}, \quad (19)$$

where:

$$\begin{aligned} A_0 &= 2 * l_2 (x_4 - x_2), \\ B_0 &= 2 * l_2 (y_4 - y_2), \\ l_{24} &= \sqrt{(x_4 - x_2)^2 + (y_4 - y_2)^2}, \end{aligned}$$

the leg length l_0 and tilt angle θ can be analytically derived from the kinematic relationships established in Fig. 6 through algebraic manipulation:

$$\begin{aligned} l_0 &= \sqrt{(y_3 - y_0)^2 + (x_3 - x_0)^2}, \\ \theta &= \theta_0 - \frac{\pi}{2}. \end{aligned} \quad (20)$$

3.2.2 Adaptive LQR

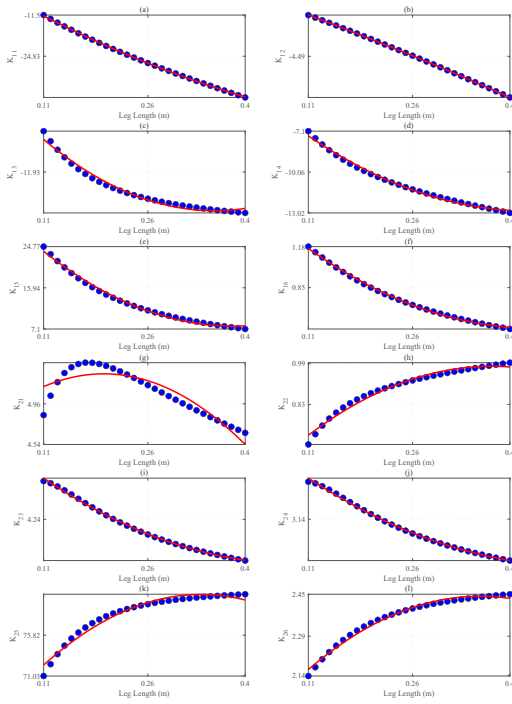


Fig. 7: K Matrix Quadratic Term Fitting.

The LQR gain matrix $\mathbf{K}$ is systematically sampled at 5 mm increments across the leg length range $L \in [0.1, 0.4]$ m. For each element K_{ij} in the gain matrix, a quadratic polynomial regression model is established through MATLAB's Curve Fitting Toolbox[11]:

$$K_{ij}(L) = \alpha_{ij}L^2 + \beta_{ij}L + \gamma_{ij}. \quad (21)$$

3.2.3 Simulation verification

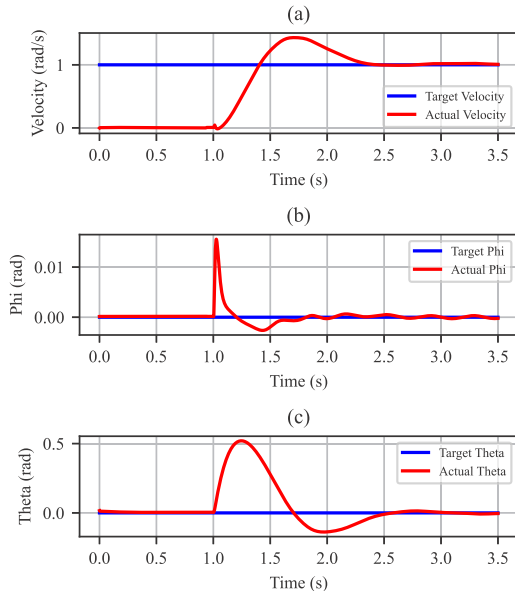


Fig. 8: Adaptive LQR Performance When $l_0 = 0.12m$.

As evidenced in Figs. 8 and 9, the proposed adaptive LQR controller exhibits robust balancing control performance for

the wheel-legged self-balancing robot under variable leg-length conditions, thereby satisfying all predefined control requirements and design specifications. The simultaneous stabilization of the robotic chassis and rapid trajectory tracking (steady-state error < 0.05 rad/s) demonstrably validates the unique capability of variable leg-length wheel-legged robots to reconcile dynamic agility with static stability.

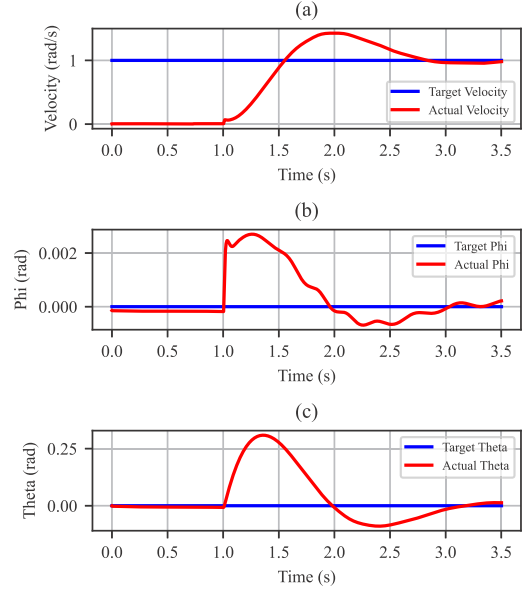


Fig. 9: Adaptive LQR Performance When $l_0 = 0.32m$.

4 Integrated Motion Control

4.1 Leg Length Control

The single-leg length control is implemented through a hybrid PID-feedforward strategy. Based on the quasi-static force distribution analysis of the five-link mechanism (Fig. 6)[17], the PID-generated corrective torque T_{PID} and LQR-derived feedforward torque T_p are strategically integrated to compute the reference joint torque vector. This fused torque command ensures precise length stabilization while maintaining kinematic consistency across leg configurations. In the context of bipedal balance control, this study employs roll angle error as the primary control variable for proportional-integral-derivative (PID) regulation to ensure zero-roll equilibrium during dynamic leg length adjustments. The proposed control strategy dynamically modulates the compensatory torque distribution between individual legs through a feedback loop that minimizes the deviation between the measured roll angle and the desired reference (0 rad). To enhance robustness against parametric uncertainties arising from leg length variations, a sensor fusion framework integrating inertial measurement unit (IMU) data with joint encoder measurements is employed to ensure accurate error estimation. This approach decouples the kinematic disturbances caused by asymmetric leg configurations from the body attitude control loop, enabling real-time stabilization of the roll dynamics while preserving energy efficiency through torque optimization.

4.2 Steering Control

While LQR control demonstrates superior transient performance for yaw angle regulation in wheel-legged robots, the relative simplicity of steering dynamics renders PID control sufficient for practical implementation[18]. Through PID-based yaw angle feedback, precise heading stabilization and smooth trajectory tracking are achieved without incurring the computational overhead associated with LQR synthesis[19]. Furthermore, the robustness of PID controllers to parameter uncertainties and external disturbances—common in unstructured environments—further justifies their preference for baseline steering control. While LQR offers optimal performance under precise system models, its reliance on state-space linearization and quadratic cost function tuning introduces sensitivity to model inaccuracies, which may outweigh its benefits in low-complexity steering tasks. This trade-off analysis substantiates PID control as a baseline strategy for steering tasks where model fidelity cannot be guaranteed and computational parsimony is prioritized over theoretical optimality. The controller's asymptotic stability and disturbance rejection capabilities are particularly advantageous in unstructured environments requiring adaptive navigation.

5 Conclusion

This paper establishes an adaptive control framework for wheel-legged self-balancing robots through systematic dynamic modeling and controller synthesis. The double inverted pendulum abstraction effectively captures essential nonlinear dynamics while enabling tractable linearization for real-time implementation. By integrating gain-scheduled LQR with virtual model control principles, the proposed method achieves full-state stabilization across variable leg-length configurations (0.1–0.4 m). Compared to fixed-gain LQR implementations, the adaptive architecture shows superior robustness against parametric variations induced by leg adjustments, while maintaining computational efficiency suitable for embedded deployment. This work provides a foundation for developing terrain-adaptive wheel-legged systems, with future extensions focusing on hardware validation and multi-modal locomotion integration.

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